

Probability Theory: Expectation, Inequalities, and Convergence

Based on Chapters 2–3 Time: 150 minutes Total: 100 points

Instructions: Answer all questions in the spaces provided. Show all steps for computational problems. Justify your reasoning for conceptual and case study questions.

Part I: Multiple Choice (12 points)

Choose the best answer for each question. (2 points each)

1. Which of the following statements about expectation is **always** true?
 - A. $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ for any random variables X, Y .
 - B. $\mathbb{E}(g(X)) = g(\mathbb{E}(X))$ for any function g .
 - C. $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$ for constants a, b .
 - D. $\mathbb{E}(X^2) = (\mathbb{E}(X))^2$ for any X .
2. Let X be a random variable with finite variance. Chebyshev's inequality states:
 - A. $\mathbb{P}(|X - \mathbb{E}(X)| \geq t) \leq \frac{\mathbb{V}(X)}{t^2}$ for all $t > 0$.
 - B. $\mathbb{P}(|X - \mathbb{E}(X)| \geq t) \geq \frac{\mathbb{V}(X)}{t^2}$ for all $t > 0$.
 - C. $\mathbb{P}(|X| \geq t) \leq \frac{\mathbb{E}(|X|)}{t}$ for all $t > 0$.
 - D. $\mathbb{P}(|X - \mathbb{E}(X)| \geq t) \leq \frac{\mathbb{E}(|X|)}{t}$ for all $t > 0$.
3. If $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$, which of the following is **not** necessarily true?
 - A. $X_n + Y_n \xrightarrow{P} X + Y$.
 - B. $X_n Y_n \xrightarrow{P} XY$.
 - C. $\mathbb{E}(X_n) \rightarrow \mathbb{E}(X)$.
 - D. $g(X_n) \xrightarrow{P} g(X)$ for any continuous function g .
4. The moment generating function (MGF) of a random variable X is defined as $\psi_X(t) = \mathbb{E}(e^{tX})$. Which statement about MGFs is correct?
 - A. The MGF always exists for all real t .
 - B. If $\psi_X(t) = \psi_Y(t)$ for all t in an open interval containing 0, then X and Y have the same distribution.
 - C. $\psi_X^{(j)}(0) = j! \mathbb{E}(X^j)$.
 - D. MGFs are only defined for discrete random variables.
5. For a convex function g , Jensen's inequality states:
 - A. $\mathbb{E}(g(X)) \leq g(\mathbb{E}(X))$.
 - B. $\mathbb{E}(g(X)) \geq g(\mathbb{E}(X))$.
 - C. $\mathbb{E}(g(X)) = g(\mathbb{E}(X))$ if g is linear.
 - D. Both (b) and (c) are correct.
6. Let $X_1, \dots, X_n$ be i.i.d. with mean μ and finite variance σ^2 . The Central Limit Theorem says:
 - A. $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{D} N(0, 1)$.
 - B. $\bar{X}_n \xrightarrow{P} \mu$.
 - C. $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{D} N(0, \sigma^2)$.
 - D. Both (a) and (c) are correct.

Part II: Fill in the Blanks (10 points)

Complete each statement with the correct expression. (2 points each blank)

1. For any random variable X with finite mean μ_X , the variance can be computed as $\mathbb{V}(X) = \mathbb{E}(X^2) - \underline{\hspace{2cm}}$.
2. If X and Y are independent, then $\text{cov}(X, Y) = \underline{\hspace{2cm}}$ and $\rho_{X,Y} = \underline{\hspace{2cm}}$.
3. The weak law of large numbers (version 2) states that if $X_1, \dots, X_n$ are i.i.d. with finite mean μ , then $\bar{X}_n \xrightarrow{P} \mu$. This is a statement about convergence in $\underline{\hspace{2cm}}$.
4. For a non-negative random variable X with finite expectation, Markov's inequality gives $\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}(X)}{t}$ for any $t > 0$. Applying this to $|X - \mu_X|^2$ yields Chebyshev's inequality: $\mathbb{P}(|X - \mu_X| \geq t) \leq \underline{\hspace{2cm}}$.
5. If $X_1, \dots, X_n$ are i.i.d. with finite mean μ and finite variance σ^2 , then the Delta method with $g(\mu) = e^\mu$ gives that $\sqrt{n}(e^{\bar{X}_n} - e^\mu)$ is approximately distributed as $N(0, \underline{\hspace{2cm}})$.

Part III: Short Answer (18 points)

Provide concise but complete answers. (6 points each)

1. (Properties of expectation and variance)

- (a) State the linearity of expectation: $\mathbb{E}(\sum_{i=1}^n a_i X_i) = \underline{\hspace{2cm}}$.
- (b) Prove that $\mathbb{V}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$ using the definition $\mathbb{V}(X) = \mathbb{E}[(X - \mu_X)^2]$.
- (c) If X_1 and X_2 are independent, show that $\mathbb{V}(X_1 + X_2) = \mathbb{V}(X_1) + \mathbb{V}(X_2)$.

2. (Modes of convergence) Define the following three types of convergence for a sequence of random variables X_n and a random variable X :

- (a) $X_n \xrightarrow{P} X$ (convergence in probability)
- (b) $X_n \xrightarrow{D} X$ (convergence in distribution)
- (c) $X_n \xrightarrow{qm} X$ (convergence in quadratic mean)

State the logical implications among them (e.g., $A \Rightarrow B$). Provide a counterexample to show that convergence in probability does not imply convergence in quadratic mean.

3. (Moment generating functions)

- (a) Define the MGF $\psi_X(t)$ of a random variable X . What is the relationship between $\psi_X^{(j)}(0)$ and the j -th moment $\mathbb{E}(X^j)$?
- (b) Suppose $X \sim \text{Poisson}(\theta)$. Find its MGF $\psi_X(t)$.
- (c) Use the MGF to compute $\mathbb{E}(X)$ and $\mathbb{V}(X)$ for the Poisson distribution.

Part IV: Calculation Problems (40 points)

Show all steps.

1. (8 points) (Expectation and variance of common distributions)

- (a) Let $X \sim \text{Binomial}(n = 12, \theta = 0.4)$. Compute $\mathbb{E}(X)$ and $\mathbb{V}(X)$.
- (b) Let $Y \sim \text{Exponential}(\theta = 3)$ (rate parameter). Find $\mathbb{E}(Y)$ and $\mathbb{V}(Y)$.
- (c) Let $Z \sim \text{Gamma}(a = 2, b = 4)$ (shape a , scale b). Compute $\mathbb{E}(Z)$ and $\mathbb{V}(Z)$.

(d) If $W \sim N(\mu = 3, \sigma^2 = 16)$, find $\mathbb{E}(W^2)$.

2. (8 points) (Applying probabilistic inequalities)

- (a) A non-negative random variable X has $\mathbb{E}(X) = 8$. Use Markov's inequality to bound $\mathbb{P}(X \geq 20)$.
- (b) A random variable Y has mean $\mu = 5$ and variance $\sigma^2 = 0.36$. Use Chebyshev's inequality to find an upper bound for $\mathbb{P}(|Y - 5| \geq 1.2)$.
- (c) Suppose U and V have finite second moments with $\mathbb{E}(U^2) = 9$ and $\mathbb{E}(V^2) = 16$. Apply the Cauchy-Schwarz inequality to bound $\mathbb{E}(|UV|)$.
- (d) For a convex function $g(x) = x^2$ and a random variable X with $\mathbb{E}(X) = 2$, use Jensen's inequality to compare $\mathbb{E}(X^2)$ and $(\mathbb{E}X)^2 = 4$.

3. (8 points) (Conditional expectation and variance) Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2e^{-x}e^{-y}, & 0 < x < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the marginal PDF $f_Y(y)$ and the conditional PDF $f_{X|Y}(x|y)$ for $y > 0$.
 - (b) Compute $\mathbb{E}(X|Y = y)$.
 - (c) Use the law of total expectation to find $\mathbb{E}(X)$.
 - (d) Compute $\mathbb{V}(X)$ using the law of total variance: $\mathbb{V}(X) = \mathbb{E}[\mathbb{V}(X|Y)] + \mathbb{V}[\mathbb{E}(X|Y)]$.
- 4. (8 points) (Convergence and limit theorems)** Let $X_1, X_2, \dots$ be i.i.d. random variables with mean $\mu = 2$ and variance $\sigma^2 = 9$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- (a) State the Weak Law of Large Numbers (WLLN). What does it tell us about $\bar{X}_n$ as $n \rightarrow \infty$?
 - (b) For $n = 100$, approximate $\mathbb{P}(1.8 \leq \bar{X}_{100} \leq 2.2)$ using the Central Limit Theorem.
 - (c) Suppose we define $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$. What does S_n^2 converge to in probability? Justify briefly.
 - (d) According to Theorem 3.6 (continuous mapping), if g is continuous, then $g(\bar{X}_n) \xrightarrow{P} g(2)$. Use this to find the limit of $e^{\bar{X}_n}$ as $n \rightarrow \infty$.
- 5. (8 points) (Delta method)** Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}(X_i) = 5$ and $\mathbb{V}(X_i) = 4$. Define $Y_n = \bar{X}_n^2$.
- (a) Use the delta method with $g(x) = x^2$ to find the approximate distribution of $\sqrt{n}(Y_n - 25)$ for large n . Specify the asymptotic mean and variance.
 - (b) If $g(x) = \sqrt{x}$, what is the approximate distribution of $\sqrt{n}(\sqrt{\bar{X}_n} - \sqrt{5})$? (Assume $g'(5) \neq 0$.)

Part V: Case Studies (20 points)

- 1. (10 points) (Insurance risk analysis)** An insurance company models the number of claims N per day as a Poisson random variable with mean $\lambda = 20$. The claim amount X_i for each claim is independent of N and has an exponential distribution with mean \$2000 (i.e., rate $1/2000$). The total daily claim amount is $S = \sum_{i=1}^N X_i$ (with $S = 0$ if $N = 0$).
- (a) Use the law of total expectation to compute $\mathbb{E}(S)$.
 - (b) Use the law of total variance: $\mathbb{V}(S) = \mathbb{E}[\mathbb{V}(S|N)] + \mathbb{V}[\mathbb{E}(S|N)]$ to compute $\mathbb{V}(S)$. (Recall: for fixed $N = n$, S is sum of n i.i.d. exponentials.)

- (c) The company wants to estimate the probability that S exceeds \$50,000. Use Chebyshev's inequality to obtain an upper bound for $\mathbb{P}(S > 50000)$ based on your computed mean and variance.
- (d) Over 100 independent days, let $\bar{S}_{100}$ be the sample mean of daily total claims. Approximate $\mathbb{P}(\bar{S}_{100} > 42000)$ using the Central Limit Theorem (assume the true variance $\mathbb{V}(S)$ is known from part (b)).
- 2. (10 points) (Convergence of estimators)** Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}(X_i) = \mu$ and $\mathbb{V}(X_i) = \sigma^2 < \infty$. Consider two estimators for μ :

$$\hat{\mu}_n^{(1)} = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\mu}_n^{(2)} = \frac{1}{n+1} \sum_{i=1}^n X_i.$$

- (a) Compute the bias $\text{Bias}(\hat{\mu}_n^{(2)}) = \mathbb{E}(\hat{\mu}_n^{(2)}) - \mu$.
- (b) Show that both estimators converge in probability to μ as $n \rightarrow \infty$. Which one converges in quadratic mean? Justify.
- (c) Find the asymptotic distribution of $\sqrt{n}(\hat{\mu}_n^{(1)} - \mu)$ and $\sqrt{n}(\hat{\mu}_n^{(2)} - \mu)$ (if they converge). If they do not converge, explain why.
- (d) Suppose you are told that $\sigma^2 = 1$ and $n = 100$. Approximate $\mathbb{P}(|\hat{\mu}_n^{(2)} - \mu| < 0.2)$ using the CLT, accounting for the bias.

End of Examination